

Comprehensive Interview Q&A – Alpha Signal Project

Explain your project in 1–2 minutes

My project focused on analyzing historical stock price data of Apple Inc. to identify statistical patterns and market inefficiencies. I performed data cleaning, exploratory data analysis, and feature engineering by computing log returns, rolling volatility, and momentum indicators. Based on these features, I constructed alpha signals using momentum and volatility rules. I then backtested these signals against a buy-and-hold benchmark. The results showed that while the strategy underperformed in total returns, it achieved better risk-adjusted performance due to lower volatility.

What was the objective of your analysis?

The objective was to test whether simple rule-based strategies using momentum and volatility can generate alpha, i.e., returns higher than a benchmark like buy-and-hold.

Why did you choose stock data (Apple Inc.)?

Apple Inc. is a large-cap stock with high liquidity and rich historical data, making it ideal for time-series analysis and testing financial models.

What problem were you trying to solve?

I aimed to identify whether market inefficiencies exist that can be exploited using simple quantitative signals.

What is an alpha signal?

An alpha signal is a predictive rule or indicator designed to generate excess returns relative to a benchmark by exploiting patterns in the data.

What is your main finding?

The strategy reduced volatility and improved Sharpe ratio but underperformed buy-and-hold in total returns.

What did you observe from the price trend?

The price trend showed a strong upward movement with occasional sharp declines, indicating long-term growth with short-term volatility.

What does the returns distribution tell you?

It shows that returns are centered around zero with fat tails, indicating frequent extreme events.

What are fat tails?

Fat tails refer to the higher probability of extreme outcomes compared to a normal distribution.

Why are financial returns not normally distributed?

Due to market shocks, investor behavior, and external events, returns exhibit skewness and kurtosis.

What is volatility clustering?

It is the phenomenon where high volatility periods are followed by high volatility.

What did your volatility plot show?

It showed clear clustering of volatility, confirming persistence in market risk.

Why is EDA important before modeling?

It helps understand the data, detect anomalies, and validate assumptions before building models.

What is the difference between simple return and log return?

Simple return is percentage change, while log return is the logarithmic change. Log returns are additive over time.

Why did you use log returns?

They simplify calculations and are better suited for statistical modeling.

What does mean return represent?

It represents the average daily return.

Why is mean return close to zero?

Because markets are largely efficient and unpredictable in the short term.

What does standard deviation represent in finance?

It represents volatility or risk.

What is risk in financial data?

Risk refers to uncertainty in returns, often measured by volatility.

What is compounding?

It is the process where returns generate additional returns over time.

What is autocorrelation?

It measures the relationship between a variable and its lagged values.

What did you observe about autocorrelation in your data?

Returns showed weak autocorrelation, while squared returns showed strong autocorrelation.

What is volatility clustering and how did you test it?

It is persistence in volatility, tested using autocorrelation of squared returns.

What is momentum persistence?

It is the continuation of past returns into the future.

Did your data show momentum or mean reversion?

It showed weak mean reversion.

What are stylized facts of financial markets?

They are common empirical properties like fat tails and volatility clustering.

Why did you compute rolling volatility?

To capture changing risk levels over time.

Why did you choose a 20-day window?

It approximates one trading month.

What is momentum and how did you calculate it?

Momentum measures trend strength, calculated as past returns.

Why use multiple momentum windows?

To capture both short-term and medium-term trends.

How do these features help in prediction?

They capture patterns in returns and risk.

How did you construct your alpha signals?

Using rules: buy when momentum is positive and volatility is low.

Why combine momentum and volatility?

To improve signal quality by filtering risky trades.

What is the intuition behind your strategy?

Trends persist but high volatility increases risk.

Why use a volatility filter?

To avoid unstable market periods.

What is signal threshold and how did you choose it?

I used median volatility as a threshold.

What happens if the threshold is too strict?

It reduces trading frequency and may miss profitable opportunities.

What is backtesting?

Testing a strategy on historical data.

Why is backtesting important?

It evaluates strategy performance before real implementation.

What is look-ahead bias?

Using future data in decision-making.

Why did you shift the signal?

To ensure decisions are based only on past information.

What happens if you don't shift?

It results in unrealistic performance.

What is your benchmark and why?

Buy-and-hold, because it represents passive investing.

What is buy-and-hold strategy?

Investing and holding without trading.

What metrics did you use?

Mean return, volatility, Sharpe ratio, cumulative return.

What is Sharpe ratio?

Return per unit of risk.

Why is Sharpe ratio important?

It measures risk-adjusted performance.

Why did your strategy underperform?

It avoided high-volatility periods where returns were high.

Why did it outperform in Sharpe?

Because it reduced volatility significantly.

What is risk-adjusted return?

Return adjusted for risk taken.

What is the trade-off between risk and return?

Higher returns usually come with higher risk.

Why did your strategy fail to beat buy-and-hold?

Due to excessive filtering of high-return periods.

What are the limitations?

Single stock, no transaction costs, static thresholds.

How would you improve your strategy?

Use dynamic thresholds and include more features.

What assumptions did you make?

Markets follow historical patterns.

What would happen in a bear market?

The strategy may perform better due to risk control.

Why might momentum not work?

Due to mean reversion or noisy data.

How did you clean your data?

Removed extra rows, fixed columns, converted types.

What is iloc vs loc?

iloc is position-based, loc is label-based.

Why reset index?

To maintain sequential order.

What does shift() do?

Moves data to avoid look-ahead bias.

What is rolling window?

A moving window for calculations.

How did you compute returns?

Using log returns formula.

How do you handle missing values?

Drop or fill them depending on context.

How would you include transaction costs?

Subtract cost per trade from returns.

How would you test robustness?

By varying parameters and testing across datasets.

How would you scale to multiple stocks?

Apply same logic to a portfolio.

Can ML improve this?

Yes, by capturing nonlinear relationships.

What models would you use?

Linear Regression, Random Forest, XGBoost.

How would you avoid overfitting?

Cross-validation and out-of-sample testing.

What skills did you gain?

Python, EDA, financial analysis.

What was most challenging?

Designing effective signals.

What would you do differently?

Include more advanced models.

How is this relevant?

It demonstrates quantitative and analytical skills.